Edward Stanford

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Education

M.S. Computer Science *University of Utah* 2025 - 2026 *expected*
B.S. Computer Science, Mathematics Minor *University of Utah* 2021 - 2025
Certificate in Data Science *University of Utah* 2021 - 2025
Dean's List; recipient of multiple merit-based scholarships *University of Utah* 2021 - 2025

Experience

Teaching Assistant (Computer Systems, Software Practice II, Object Oriented Prog) *University of Utah* August 2023 - Present
- Supported over 375 students by managing Piazza discussions, grading assignments, and providing detailed feedback on student submissions.
- Ran labs and held help hours to answer questions and clarify concepts.

GREAT Summer Camp Instructor (4th-6th Grade) *University of Utah* June 2024 - July 2024
- Taught robotics concepts to elementary students with hands-on activities.
- Supervised a structured learning environment for children.

Projects

Goldshelf - Fullstack Ranking System *TypeScript* July 2024 - Present
- Developed a full-stack web application for ranking and reviewing content of any user defined type. Hosted on Cloudflare with D1/R2 database/storage.
- Custom self repairing binary search insertion to maintain a sorted list of user defined content.

Full Stack Photo Sharing App (Capstone Project) - BeThere *C#, React* August 2024 – May 2025
- Developed a backend server with a modular REST API to handle photo uploads, user management, and clustering based on geolocation data.
- Integrated PostgreSQL for robust data storage with LINQ, and built access control linking users to clusters for selective album visibility by location.

Full JPL Compiler *C++, Rust* January 2025 – May 2025
- Developed a full compiler for the JPL programming language.
- Implemented the full pipeline (lexing, parsing, semantic analysis, optimization, codegen) including type checking, scope management, and code optimizations.

Path of Memories Gamejam *C# with Unity* January 2023
- Built a 2D platformer featuring a robust dialogue system, player progression tracking, and character-driven interactions/level design.
- Added diverse player abilities including wall climbing, double jumping, and dashing.

Circuit Simulator QT Application *C++* April 2023
- Designed an educational circuit-logic game with an intuitive, interactive UI and user-customizable gates.
- Implemented save functionality using JSON for persistent user data.

LMS Website *C#* April 2023
- Developed a learning management system leveraging a MariaDB backend.
- Ensured seamless front-end/back-end integration using .NET technologies, including user authentication and course management.

Sprite Editor QT Application *C++* March 2023
- Created a comprehensive sprite editor with detailed UI and user interactions.
- Supported multiple image formats/export options and implemented advanced drawing tools with color manipulation features.

Snake Network Game *C#* October 2022 – December 2022
- Built a networked snake game with distinct server and client implementations.
- Managed multiplayer functionality with real-time synchronization.

Miscellaneous

Advent of Code 2025 *Rust* December 2025 - January 2026
- Solved 22 challenges focusing on dynamic programming, graph algorithms, and geometric problems.
- Improved Rust proficiency while implementing efficient data structures/algorithms to optimize solution performance.

Project Euler *C++, Python* August 2020 – December 2020
- Tackled 59 mathematics and computation-intensive problems.
- Employed efficient algorithms for large-scale numerical challenges, strengthening mathematical reasoning and coding proficiency.

Technology

Experienced
- C++, Rust, C, C#, SQL, VSCode, Qt, Docker, Linux, MacOS

Proficient
- Python, Java, Latex, Typst, R, Git, MS Visual Studio, MS Office, XCode